Manual

Enhanced Control Charts and Histograms

WEP-ERH 8.2

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Contents

1 Enhanced Control Charts and Histograms - Overview

Purpose

This component is used when the attributive and variable characteristics of collected goods receipt inspection data have to be displayed across orders in combined control chart and histogram format.

Implementation Considerations

Since in general only one sample exists for each characteristic during goods receipt inspections, no graphical historical overview to detect developments can be displayed, if data can only be displayed for a single goods receipt inspection. This component can be used if analysis across goods receipt inspections is required. For example, analysis over a longer period can show the effect of improvements implemented by the supplier.

This component requires the inspection data to have been captured in HYDRA.

Integration

The component "Inspection Planning for Goods Receipt Inspections" with its associated inspection requirements and inspection results forms the base data, where the inspection data must have been previously entered through the AIP data entry client.

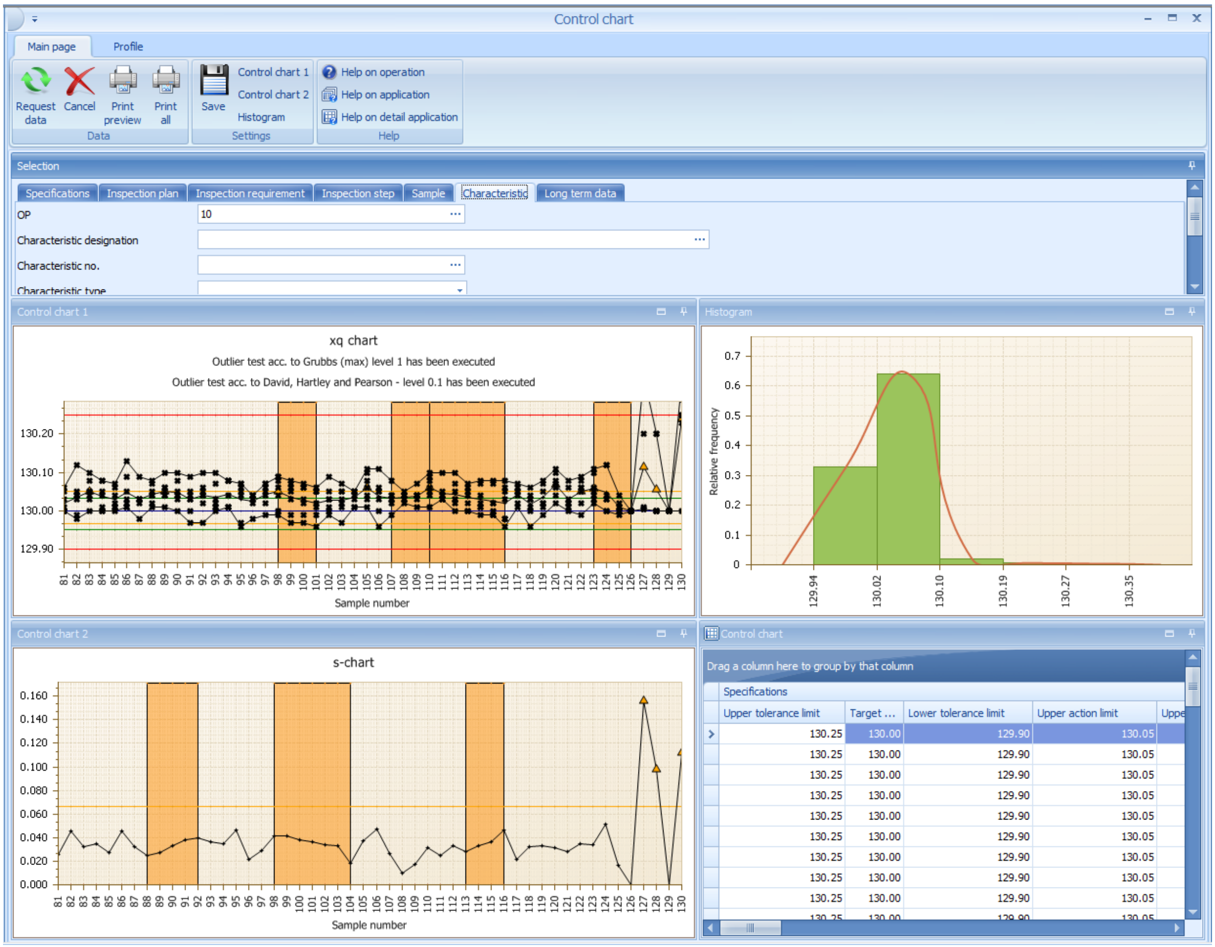
Features

Tested characteristics can be displayed graphically across orders in various control chart formats, as a histogram and in a list of the base data. The base data can be restricted through a wealth of available filters.

2 Control chart evaluation

Overview

Menu	Quality management → QM evaluation → Control chart
Transaction code	ccep
Function authorization	ccep



Purpose

The "Control chart" evaluation shows different control charts across all orders for the characteristics that have been checked. The evaluation also includes the histogram and the list of data the evaluation is based on. You can use multiple filters to narrow down the data basis.

Integration

Control chart evaluations are available in the following areas:

- Goods receipt
- Initial sample inspection
- Production
- Goods issue

The evaluations use the recorded inspection data. Therefore, the "Control chart" evaluation is a global analysis tool.

Requirements

To display control charts, you must populate the following filter fields:

- Type ("specifications" tab)
- Area ("specifications" tab)
- OP sequence or characteristic number ("characteristic" tab)

You can only identify inspection order characteristics uniquely, if you enter the operation sequence number. As the same characteristic number may be used several times within one inspection step, the inspection step characteristic cannot be identified uniquely even if the order number or the inspection requirement and inspection step number are added as further filter criteria. As the specified tolerance limits are not necessarily identical for the complete data set that has been filtered, the calculation of statistical key figures, e.g. cpk value that is printed in a form, is normally suppressed. You can only identify the inspection step characteristic uniquely if you filter by the operation sequence number, the inspection requirement number and the inspection step number.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

You can use some of the fields under specific conditions only. For example, if you filter by a cavity number, you require the license for a cavity-related collection. This is similar with inspection point fields. These fields require an inspection point related collection of inspection data.

Tab *"Specifications"*

Type

Selection list of area types

- in-production inspection
- goods receipt inspection
- goods issue inspection
- initial sample inspection

The list of area types depends on the respective licenses in use. The list can be restricted.

Area

Selection list of the configured areas of the previously filtered area type. By default, the following areas are available:

- In-production inspection: production
- Goods receipt inspection: goods receipt
- Goods issue inspection: goods issue
- Initial sample inspection: initial sample

Further areas can be generated through customizing.

Tab *"Inspection step"*

Operation

Operation number

Tab *"Inspection point identification"*

If these fields are not available, you require a new program version of this application.

For these filter fields, the data collection must be performed with reference to the inspection points. This is usually the case in production, but not in goods receipt.

Workplace

Workplace number

Partial batch

Number of partial batch

ERP batch

ERP batch number

Field 1…3

Additional fields for inspection points.

For the area "Production", *Field 1* specifies the "tool number" and *Field 3* the sample number in the default configuration.

Tab *"Additional fields for inspection points“*

If these fields are not available, you require a new program version of this application.

For these filter fields, the data collection must be performed with reference to the inspection points. This is usually the case in production, but not in goods receipt.

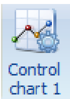
All fields included in this application do not support matchcode filtering for technical reasons.

Field 4…8

Additional fields for inspection points.

For the area "Production", *Field 4* specifies the batch in the default configuration.

Toolbar



Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The corresponding details are described in the respective detail application.

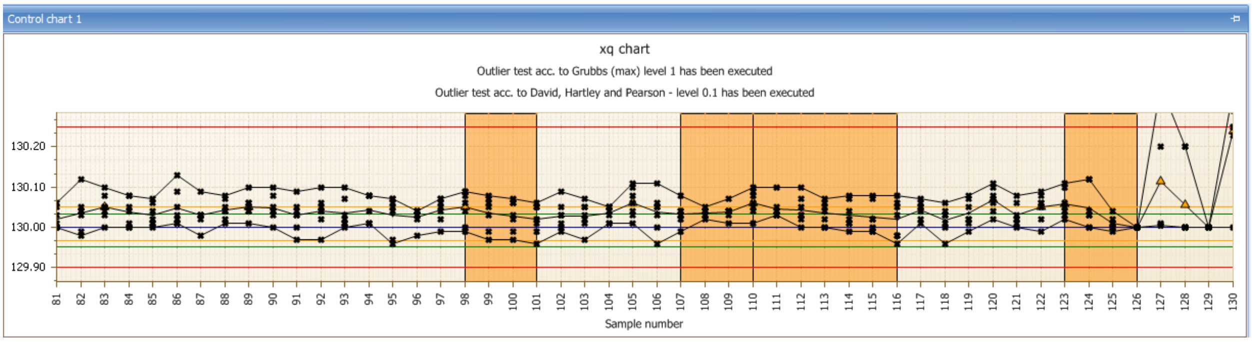
Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The corresponding details are described in the respective detail application.

Histogram settings

Opens a dialog to configure histogram settings. The corresponding details are described in the respective detail application.

"Control chart 1" detail applications



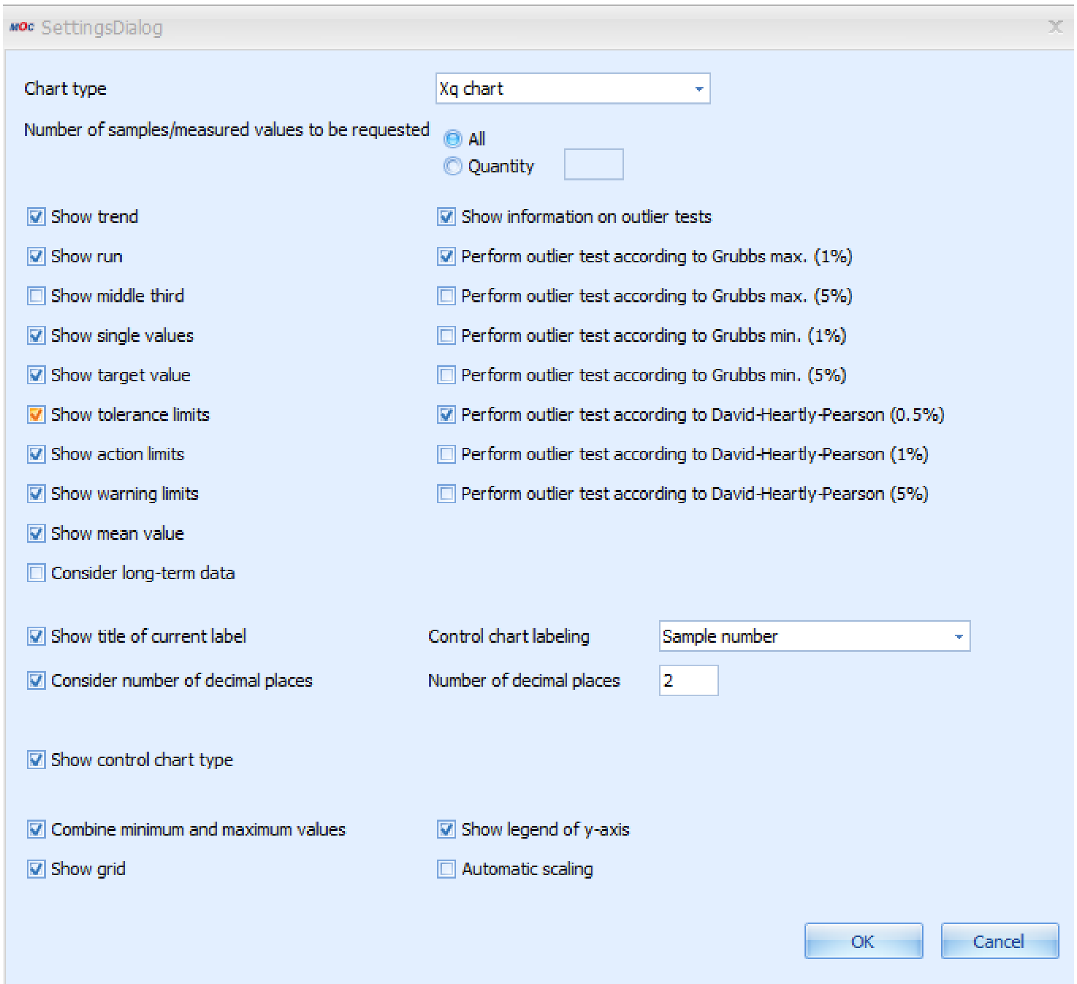
Call the dialog to configure "Control chart 1" to define the contents of this application. If you use this dialog to make changes, the changes are saved per user. If no specific setting has been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed by default. The available control charts depend on the characteristic type. By default, the user can select one of the following control charts.

Variable characteristic

- Xq chart
- S chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart



The essential configuration options are described in the paragraphs that follow.

Number of samples/measured values to be requested

Specifies the number of samples or single measured values that are displayed in the control chart.

Show …

The options that include the term "Show ..." enable or disable the display of the corresponding data in the control chart.

Consider long-term data

Check this option to integrate archived data of the medium-term data area.

Combine minimum and maximum values

It is recommended to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of variation of single values within a sample.

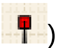
Automatic scaling

Using the "automatic scaling" function, you can display all values, irrespective of the existing limit values. This shows even extreme outliers in the control chart. Disadvantage: The other measured values are much smaller in the layout and it is more difficult to identify changing values or a trend.

Show trend / run / middle third

Use the monitoring functions "trend", "run" and "middle third" to better monitor a process. These functions are only available with the control charts xq and median. If you show the trend, you can visualize a process trend that may rise or fall. The system uses several samples to generate a trend. By default, these are seven consecutive rising or falling values. The run shows sections where the process runs above or below the mean value (when displayed, otherwise the target value). The run covers several samples. By default, these are seven consecutive values that are above the mean value. The default number of seven samples/values that is set to identify a trend or run can be changed via system customization. The system identifies a "middle third", if an unusually high or low number of values lies within the middle third of the range between the action limits.

The system automatically makes the respective analyses for "trend", "run" and "middle third". If a trend/run and/or middle third is identified, the control chart shows it in a graphic form. The following events are shown in the control chart via symbols or colors:

- Trend (colored area)
- Run (colored area))
- Middle third (colored area)
- Outlier (symbol: )
- Xq violates action limit (symbol )

The presentation of outliers is only possible in the xq chart and median chart. Outliers can only be displayed when single values are shown. The function to perform outlier tests and the function to display outliers are separated. You can activate the different outlier tests for the different levels separately. If you have enabled the display of outlier tests, the test result is shown in text form on top of the respective control chart.

The below outlier tests are available for the different inspection levels. The inspection level is specified in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the total of all samples, i.e. each sample is considered individually. Consequently, the following phenomena may appear:
    - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
    - Despite a low range between minimum and maximum value, an outlier is identified. Possible reason: a lot of values concentrate in one “point” so that a single value having a certain distance to this point is identified as outlier within the sample.
- To perform an outlier test, there must be at least three values in the sample. The larger the number of values in a sample, the more uniform is the overall picture of the outliers compared to all samples.
- The Grubbs outlier test is performed with a sample size of $2 < n < 148$ (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed with a sample size of $2 < n < 1251$ (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- ERP batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion
- Partial batch
- Workplace
- Production workplace
- Field 1
- Field 2
- Field 3

- Field 4
- Field 5
- Field 6
- Field 7
- Field 8

The fields "field 1" to "field 8" are enabled as part of an MPDV customization. The field labels are assigned individually. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document.

Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tooltips within the control chart

If the system has identified a value as outlier, the detailed information is shown when you mouse over the symbol (red rhomb). The tooltip includes the test(s) that issued this outlier result.

For mean values, the tool tip shows the value and date and time of the first and last measured value of this sample and the respective inspection requirement number and inspection step number.

For single measured values, the tooltip shows the exact value.

For the colored areas that identify a trend, run or middle third, a respective note including the reason is shown .

Sorting of measured values

The server decides on the sorting of a control chart. The sorting type depends on the settings of control chart filters. If the filters "operation sequence" and "inspection requirement number" or "operation sequence" and "inspection step number" are entered, the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or if it is filled and the fields "inspection requirement number" and "inspection step number" are left empty.

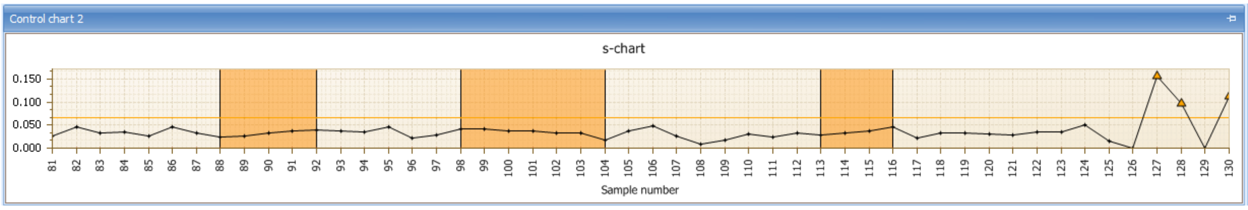
The displayed sorting of measured values influences the visual presentation of a trend or run. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers of the operation sequence, the inspection requirement and the inspection step are always known at the time data is recorded.

Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

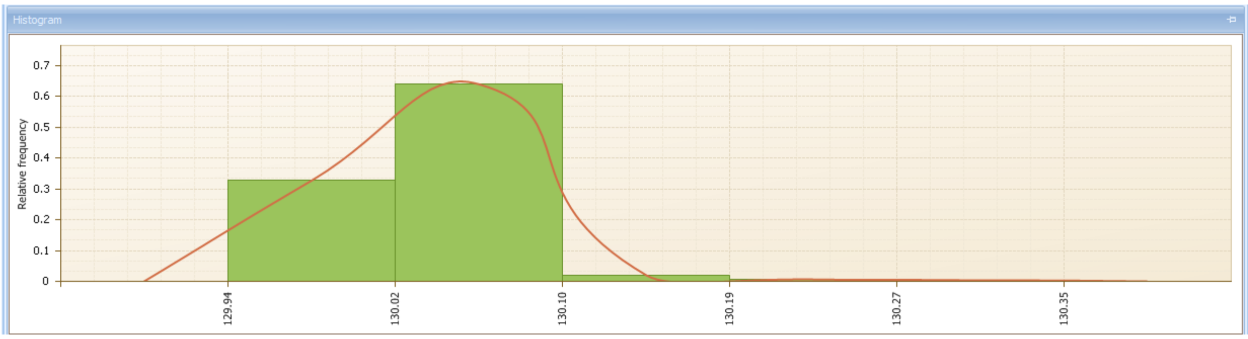
- The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

Detail applications "Control chart 2"

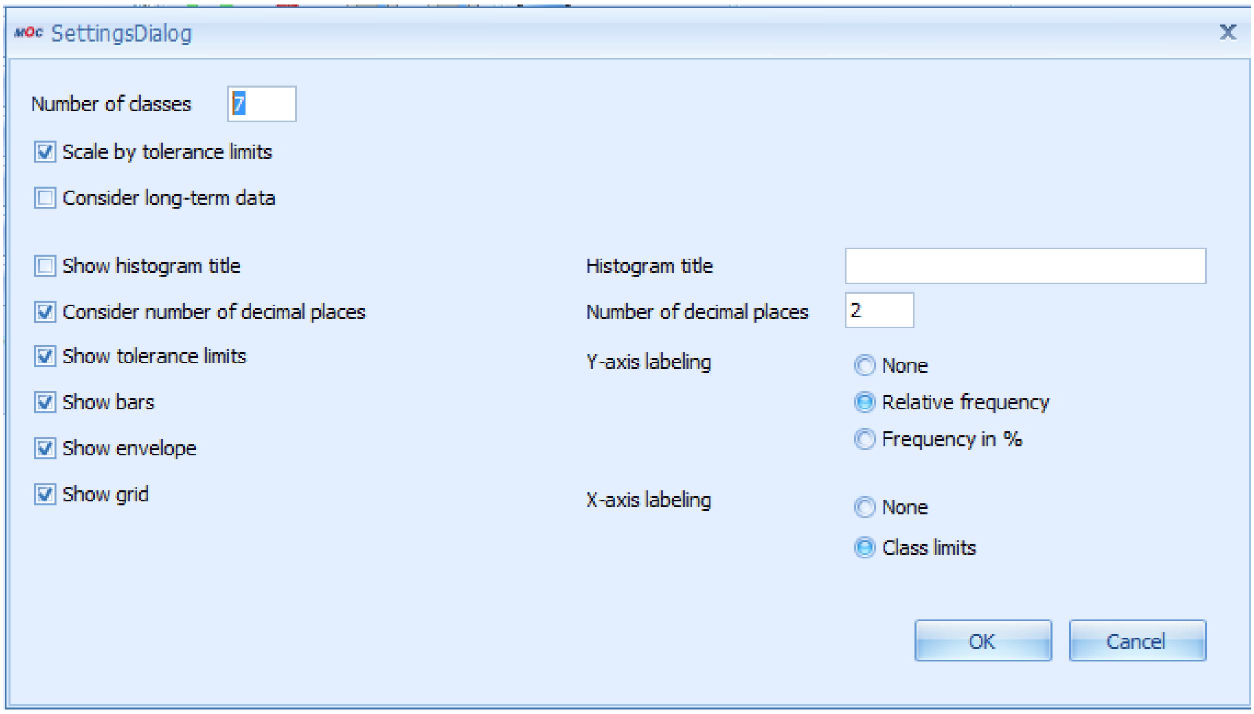


The features and configuration options of the "control chart 2" detail application are not explained here, as they match those of the "control chart 1" detail application.

Detail applications "Histogram"



You can restrict the number of samples displayed in a control chart. This is not possible with a histogram. The histogram is always based on the total of available samples matching the selection filter criteria. The number of classes and the additionally displayed information influence the histogram presentation. Call the dialog to configure the "histograms" to define the contents of this application. If you use this dialog to make changes, the changes are saved per user.



The essential configuration options are described in the paragraphs that follow.

Number of classes

Specifies the number of histogram classes. The measured values are classified into these classes to be displayed. If the histogram shows the values within the tolerance limits (option "Scale by tolerance limits"), two histogram classes include the values exceeding the tolerance limit (upper and lower).

Scale by tolerance limits

Enabled: The classes are between the tolerance limits - two "outlier classes" being displayed, one to the left and one to the right.

Disabled: The classes include the total range of all measured values (no separate classes for values exceeding the tolerance limits).

Consider long-term data

Includes the archived data from the medium-term data area.

Show histogram title

If a special title is to be displayed it may be entered by enabling this option.

X-axis labeling

If the option "Class limits" is set, the x-axis shows the respective values of the class limit. You can use the two configuration options for decimal places to specify the detail, i.e. the number of decimal places.

Consider the number of decimal places

With this option, the defined number of decimal places is used for the x-axis labeling.

Detail applications "Control chart 1 list" and "Control chart 2 list"


Control chart

Drag a column here to group by that column

Specifications								Main page		
Upper tolerance limit	Target value	Lower tolerance limit	Upper action limit	Upper warning limit	Lower action limit	Mean value	Lower warning li...	Date of first me...	Time of first me...	Date
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/13/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/11/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/10/
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/9/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			3/9/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2
130.25	130.00	129.90	130.05	130.03	129.97		129.95			4/3/2

The detail applications "control chart 1 list" and "control chart 2 list" show the data of control chart 1 and 2 in list form. If required, further analyses based on this data can be made using the Excel export function. The Excel export requires a specific license.

This list does not show the user fields of inspection points for technical reasons.

Detail application "Statistics"

The detail application "Statistics" shows the usual statistical key figures for the selected characteristic on the level of inspection step characteristics.

For variable characteristics these are

- Number of samples
- Number of measured values
- Xqq
- Minimum
- Maximum
- R
- S

- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics these are

- Number of non-conforming units
- Number of defects
- p and
- u.